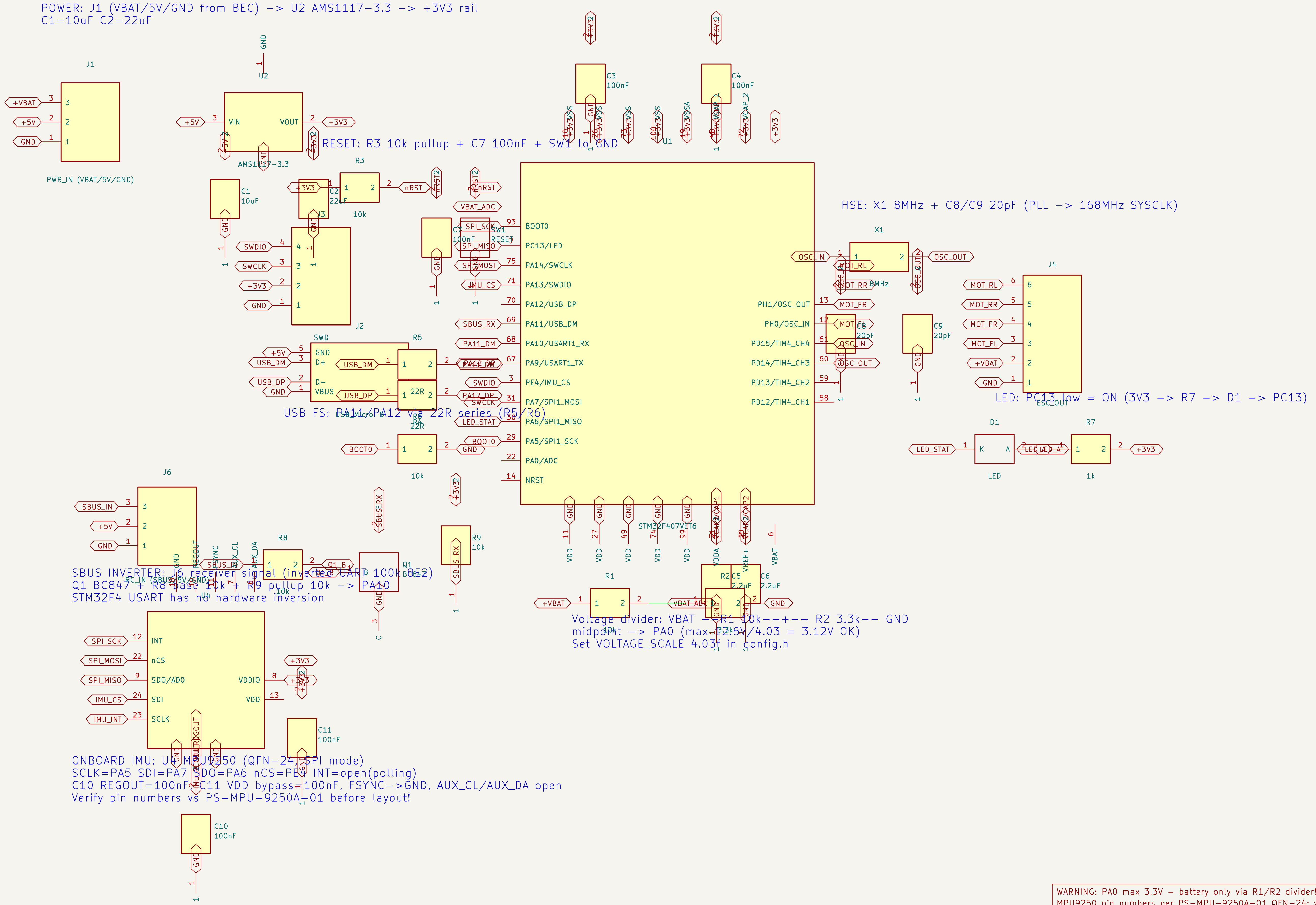

POWER: J1 (VBAT/5V/GND from BEC) -> U2 AMS1117-3.3 -> +3V3 rail
C1=10uF C2=22uF

RESET: R3 10k pullup + C7 100nF + SW1 to GND

HSE: X1 8MHz + C8/C9 20pF (PLL -> 168MHz SYSCLK)

USB FS: PA11/PA12 via 22R series (R5/R6)

LED: PC13 low = ON (3V3 -> R7 -> D1 -> PC13)

SBUS INVERTER: J6 receiver signal (inverted UART 100k 8E2)
Q1 BC847 + R8 base 10k + R9 pullup 10k -> PA10
STM32F4 USART has no hardware inversion

Voltage divider: VBAT --R1 10k--+-- R2 3.3k-- GND
midpoint -> PA0 (max 12.6V/4.03 = 3.12V OK)
Set VOLTAGE_SCALE 4.03f in config.h

ONBOARD IMU: U4 MPU9250 (QFN-24, SPI mode)
SCLK=PA5 SDI=PA7 SDO=PA6 nCS=PE4 INT=open(polling)
C10 REGOUT=100nF, C11 VDD bypass=100nF, FSYNC->GND, AUX_CL/AUX_DA open
Verify pin numbers vs PS-MPU-9250A-01 before layout!

WARNING: PA0 max 3.3V – battery only via R1/R2 divider!
MPU9250 pin numbers per PS-MPU-9250A-01 QFN-24: verify before layout
Onboard: MPU9250 (SPI), SBUS inverter (Q1), 3V3 LDO
MCU: STM32F407VET6 LQFP-100, HSE 8MHz -> 168MHz

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Title: Flix Flight Controller – STM32F407 + MPU9250 all-in-one

Size: A3	Date: 2026-10-28	**Rev: 2.0**
KiCad E.D.A. 10.0.7		Id: 1/1